Algebra 1
Unit 2
Lesson 8 Homework

Name_______Answer Key___
Date____________________
Period___________________

1. Rewrite the expression in each step using the description: $-2(x-2)+3x$

$-2x+4$	Distribute the -2 to $(x-2)$.
$-2x+4+3x$ $x+4$	Combine like terms.

2. Rewrite the expression in each step using the description: $5x+6(2x+7)$

$12x+42$	Distribute the 6 to $(2x+7)$.
$5x+12x+42$ $17x+42$	Combine like terms.

Use the order of operations to simplify each polynomial.

3. $4(x-3)+6(x-2)$

$4x-12+6x-12$
$10x-24$

4. $\frac{2}{3}(x-3)+6x$

$\frac{2}{3}x-2+6x$
$6\frac{2}{3}x-2$

5. $2(x^2 - 3x) + 4(x + 5)$

$2x^2 - 6x + 4x + 20$
$2x^2 - 2x + 20$

6. $4x^2 - 4(x - 8)$

$4x^2 - 4x + 32$

7. $(x + 2)(x - 3) - 6(x^2 + x)$

$x^2 - 3x + 2x - 6 - 6x^2 - 6x$

$-5x^2 - 7x - 6$

8. $-\frac{2}{5}(5x - 15) + 5x$

$-2x + 6 + 5x$

$3x + 6$

9. $x^2 + (x - 2)(x - 2)$

$x^2 + x^2 - 2x - 2x + 4$

$2x^2 - 4x + 4$

10. $4x(x + 1) + 2(x + 5)$

$4x^2 + 4x + 2x + 10$

$4x^2 + 6x + 10$